Computer Vision Practical Report:

DILATE AND IMAGE USING DILATE FUNCTION

1. AIM

To read an input image using OpenCV, dilate the image using the cv2.dilate() function, display both the original and dilated images side-by-side using Matplotlib, and save the dilated image to disk.

2. PROCEDURE

- Load the input image using cv2.imread('E4-IMG.png').
- Convert the image to grayscale using cv2.cvtColor().
- Create a kernel using np.ones((5,5), np.uint8).
- Apply image dilation using cv2.dilate() with the created kernel.
- Create a Matplotlib figure of size 10x5.
- Display the original and dilated images side-by-side using plt.subplot().
- Set titles as **'Original Image'** and **'Dilated Image'**.
- Disable axes using plt.axis('off').
- Display the images using plt.show().
- Save the dilated image as **dilated_image.png** using cv2.imwrite().

3. PROGRAM CODE

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

img_bgr = cv2.imread('E4-IMG.png')
img_rgb = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2RGB)

kernel = np.ones((5, 5), np.uint8)
dilated_img = cv2.dilate(img_rgb, kernel, iterations=1)

plt.figure(figsize=(12, 6))

plt.subplot(1, 2, 1)
```

```
plt.imshow(img_rgb)
plt.title('Original Image')
plt.axis('off')

plt.subplot(1, 2, 2)
plt.imshow(dilated_img)
plt.title('Dilated Image')
plt.axis('off')

plt.tight_layout()
plt.show()
```

4. OUTPUT

5. RESULT

The Python script executed successfully. The input image 'E4-IMG.png' was loaded, converted into a Canny edge image.